Panfeng Jiang, Undergraduate

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ShanghaiTech University
School of Information Science and Technology
Computer Science
Recent Works:Event-based vision, SLAM, Computer Vision
GPA:3.77/4.0 , **Ranking 8%-15%** in SIST College in the comprehensive evaluation

Research Publications

1. S. Hang, L. Tao, **Panfeng. Jiang**, G. Ling, and K. Laurent, "Led matrix-based simultaneous localization and data transmission with an event camera," *(Still Modifying)*, 2023.

- Develop the detection part and new modules for event-based localization, spatiotemporal tracking, and real-time decoding.
- Participated in the process of encoding-decoding, tracking algorithm designing and experimental testing.
- Our efforts highlighted the practical advantages of our solution by using event-based camera, particularly in achieving 64 Kbps of bandwidth and centimeter-level positioning accuracy, beyond the capabilities of CMOS cameras.

Research Experiences

2023.4 – present **Mobile Perception Lab, ShanghaiTech University, China**

Working Experiences and Social Participation

2024.11 Served as a **staff, referee** in Rubik's Cube competitions(Bay Area Speedcubin' 65 - Berkeley 2024, Berkeley Fall 2024, Berkeley October Weekday Tricubealon 2024, Cool Down Berkeley 2024.)

2023.12 Served as a **referee** in Changsha Rubik's Cube Open 2023

2023.8-11 Served as a **volunteer** in the summer camp of ShanghaiTech University , Shanghai Planetarium , Shanghai Library and the 27.5km mark of the 2023 Shanghai Marathon

2023.7 – 2023.8 Served as a **teaching assistant** in the New Oriental TOEFL course, responsible for course quality supervision and feedback, after-class homework assignments and corrections, and course answering for classmates

Ranking of the Comprehensive Evaluation

2022-2023 **Ranking 8%-15% in SIST College**

Skills

Languages English(Fluent), Mandarin Chinese(First language).

Coding Python,C/C++, …

Experimental skills Usage of ROS (Robot Operating System) , event-based camera, motion capture system.

Knowledge Algorithms and Data Structure, Machine Learning, Visual SLAM, Event-based Vision, Deep Learning (only a little).

Miscellaneous Experience

Awards and Achievements

2024.11 **Provincial Third Prize in the China Undergraduate Mathematical Contest in Modeling (CUMCM).**,

2023.12 **Department Prize for Outstanding Student Performance**, ShanghaiTech University.

Certification

CET-6 573

2023.9 **TOFEL iBT 91**

2023.3 **Certified as RHCE (Red Hat® Certified Engineer)**. Awarded by Red Hat,Inc.

Certified as RHCSA (Red Hat® Certified System Administrator). Awarded by Red Hat,Inc.

2022.12 **CET-4 642**

Relevant Course Projects

Computer Visioin

- **Disentangling Object Motion for Self-supervised Depth Estimation**(Self-supervised depth estimation heavily suffers from the mismatch of the photometric consistency caused by moving objects. And we trying to mitigate the negative impact caused by this.)

SLAM

- **Instance-level Object Pose Estimation with an Event Camera**(Event camera is able to track fast object motion, and based on this advantage, I am going to track the 6DoF object motion given the object CAD model by using convex-hull detected and ICP algorithm based methods.)
- Toy realization of Ordinary& Scale-aware & Jump-aware Pose Graph Optimization,Visual Odometry, Particle Filter and Five-point Minimal Solver for Event Camera Relative Motion Estimation.

Machine Learning

- **Use multiple models for quality prediction** (discuss the advantages and disadvantages of logistic regression, random forest, XGBoost models, and combine principal component analysis and Huber Loss for model performance optimization)

GPA in Major Courses

Overall GPA

- 3.77/4.0(Ranking in all Computer Science students:16/169,Ranking in all SIST students:27/243).

2024 Fall(Undertaking)

- Introduction to Abstract Algebra
- Introduction to Artificial Intelligence
- Efficient Algorithms and Intractable Problems
- Large Language Model Agents

2024 Spring

- **Computer Architecture I and Lab**(A)
- **Simultaneous Localization and Mapping**(A)
- **Introduction to Machine Learning**(A+)

2023 Fall

- **Algorithm and Data Structure**(A+)
- **Computer Vision I**(A)

2023 Spring

- **Discrete Mathematics**(A)
- **Mathematical Analysis II**(A)

2022 Fall

- **Introduction to Information Science and Technology**(A-)
- **Mathematical Analysis I**(A)
- **Linear Algebra I**(A-)